

Miles Smith

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Research Interests

My research sits at the intersection of biomechanics and human motor control, with a focus on understanding how the nervous system structures movement to accomplish complex tasks. I combine computational modeling with experiments to study neural control strategies in locomotion, with applications to clinical assessment and health monitoring.

Education

- 2023 – Present **Massachusetts Institute of Technology** – Cambridge, Massachusetts
SM/PhD in Mechanical Engineering
Advisor: Brian Anthony
SM Thesis: *Computational Neuromechanics of Impaired Locomotion: Using Predictive Neuromusculoskeletal Simulations to Unravel the Structure of Variability*
- 2020 – 2022 **Stanford University** – Stanford, California
MS in Civil and Environmental Engineering (Atmosphere/Energy)
- 2016 – 2020 **University of Maryland, Baltimore County** – Baltimore, Maryland
BS in Mechanical Engineering

Work Experience

- 2023 – Present **Graduate research assistant at MIT**
Advisor: Dr. Brian Anthony
- 2022 – 2023 **Mechanical engineer at Otherlab/Copper Home**
Supervisor: Dr. Samuel Calisch
- 2021 – 2022 **GEM research fellow at MIT Lincoln Laboratory (Group 47: Energy Systems)**
Supervisor: Dr. Theodore Bloomstein

Honors and Scholarships

- 2024 Wunsch Foundation Silent Hoist and Crane Award (MIT)
- 2021 USTFCCCA NCAA Division I Men's Track & Field All-Academic Award (Stanford)

- 2020 GEM Full Fellowship (Stanford)
- 2020 Arthur Ashe Jr. Sports Scholar Award (UMBC)
- 2016 Meyerhoff Premier Scholarship Award (UMBC)

Selected Publications

For a full list of publications, please see [Google Scholar](#).

- 2026 **Neuromuscular impairments alter energetic cost landscape curvature and stride speed variability in post-stroke locomotion**
Smith, Miles; Namburi, Praneeth; Seethapathi, Nidhi; Anthony, Brian
PLOS Computational Biology (Submitted for publication.)

Patents

- 2025 Battery-integrated appliance system and method (US20250071863A1)
- 2025 Battery-integrated water heater system and method (US20250075943A1)
- 2024 Induction Heating Adapter System and Method (US20240206023A1)

Teaching, Mentorship, and Service

- 2026 – Present Independent research mentor for high school students interested in sensorimotor control
- 2025 – Present Founding vice president of Melanated MechEs (MIT)
- Fall 2025 Lab instructor, 2.00: Introduction to Design (MIT)

Conferences and Workshops

- March 2026 A Robust, High-Throughput Framework for Predictive Simulations Using *MocoStudy*
OpenSim+ Advanced Workshop

Technical Skills

Programming languages: MATLAB, Python, C, C++, JavaScript, HTML/CSS

Software: ~~TeX~~ LaTeX, Git, OpenSim, SolidWorks, AutoDesk Fusion

Hardware: ESP-32, Arduino